cameras, as well as cameras in laptops and smartphones, have begun to routinely collect and analyze data on the movements of our eyes, including tiny changes to our pupils and irises lasting just a few milliseconds. Human agents are barely capable of even noticing such data, but computers can use it to calculate the direction of our gaze, based on the shape of our pupils and irises and on the patterns of light they reflect. Similar methods can determine whether our eyes are fixating on a stable target, pursuing a moving target or wandering around more haphazardly.

From certain patterns of eye movements, computers can then distinguish, for example, moments of awareness from moments of distraction, and detail-oriented people from those who pay more attention to context. Computers could infer from our eyes many additional personality traits, like how open we are to new experiences, and estimate our level of expertise in various fields ranging from reading to surgery. Experts possessing well-honed strategies display systematic gaze patterns, whereas the eyes of novices wander aimlessly. Eye patterns also indicate our levels of interest in the objects and situations we encounter, and distinguish between positive, neutral and negative interest. From this, it is possible to deduce our preferences in fields ranging from politics to sex. Much can also be known about our medical condition and our use of various substances. The consumption of alcohol and drugs – even at nonintoxicating levels – has measurable effects on eye and gaze properties, such as changes in pupil size and an impaired ability to fixate on moving objects. A digital bureaucracy may use all that information for benign purposes – such as by providing early detection for people suffering from drug abuse and mental illnesses. But it could obviously also form the foundations of the most intrusive totalitarian regimes in history.[15]

In theory, the dictators of the future could get their computer network to go much deeper than just watching our eyes. If the network wants to know our political views, personality traits and sexual orientation, it could monitor processes inside our hearts and brains. The necessary biometric technology is already being developed by some governments and companies, like Elon Musk’s Neuralink. Musk’s company has conducted experiments on live rats, sheep, pigs and monkeys, implanting electrical probes into their brains. Each probe contains up to 3,072 electrodes capable of identifying electrical signals and potentially transmitting signals to the brain. In 2023, Neuralink received approval from US authorities to begin